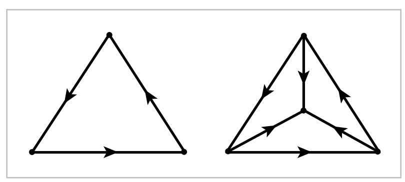
Among other things, this last condition rules out trivialities like regarding all the points of X as individual vertices. The earlier decompositions of the torus, projective plane, and Klein bottle into two triangles, three edges, and one or two vertices define Δ-complex structures with a total of six σ_α's for the torus and Klein bottle and seven for the projective plane. The orientations on the edges in the pictures are compatible with a unique ordering of the vertices of each simplex, and these orderings determine the maps σ_α.

A consequence of (iii) is that X can be built as a quotient space of a collection of disjoint simplices Δ^n_α, one for each $\sigma_\alpha : \Delta^n \to X$, the quotient space obtained by identifying each face of a Δ^n_α with the Δ^{n-1}_β corresponding to the restriction σ_β of σ_α to the face in question, as in condition (ii). One can think of building the quotient space inductively, starting with a discrete set of vertices, then attaching edges to these to produce a graph, then attaching 2-simplices to the graph, and so on. From this viewpoint we see that the data specifying a Δ-complex can be described purely combinatorially as collections of n-simplices Δ^n_α for each n together with functions associating to each face of each n-simplex Δ^n_α an $(n-1)$-simplex Δ^{n-1}_β.

More generally, Δ-complexes can be built from collections of disjoint simplices by identifying various subsimplices spanned by subsets of the vertices, where the identifications are performed using the canonical linear homeomorphisms that preserve the orderings of the vertices. The earlier Δ-complex structures on a torus, projective plane, or Klein bottle can be obtained in this way, by identifying pairs of edges of two 2-simplices. If one starts with a single 2-simplex and identifies all three edges to a single edge, preserving the orientations given by the ordering of the vertices, this produces a Δ-complex known as the 'dunce cap.' By contrast, if the three edges of a 2-simplex are identified preserving a cyclic orientation of the three edges, as in the first figure at the right, this does not produce a Δ-complex structure, although if the 2-simplex is subdivided into three smaller 2-simplices about a central vertex, then one does obtain a Δ-complex structure on the quotient space.

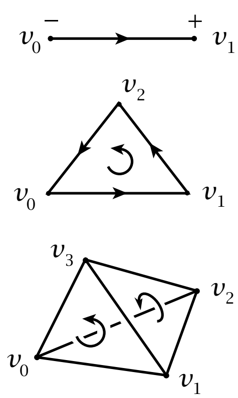


Thinking of a Δ-complex X as a quotient space of a collection of disjoint simplices, it is not hard to see that X must be a Hausdorff space. Condition (iii) then implies that each restriction $\sigma_\alpha | \mathring{\Delta}^n$ is a homeomorphism onto its image, which is thus an open simplex in X. It follows from Proposition A.2 in the Appendix that these open simplices $\sigma_\alpha(\mathring{\Delta}^n)$ are the cells e^n_α of a CW complex structure on X with the σ_α's as characteristic maps. We will not need this fact at present, however.

Simplicial Homology

Our goal now is to define the simplicial homology groups of a Δ-complex X. Let $\Delta_n(X)$ be the free abelian group with basis the open n-simplices e^n_α of X. Elements of $\Delta_n(X)$, called ***n*-chains**, can be written as finite formal sums $\sum_\alpha n_\alpha e_\alpha^n$ with coefficients $n_\alpha \in \mathbb{Z}$. Equivalently, we could write $\sum_\alpha n_\alpha \sigma_\alpha$ where $\sigma_\alpha : \Delta^n \to X$ is the characteristic map of e_α^n, with image the closure of e_α^n as described above. Such a sum $\sum_\alpha n_\alpha \sigma_\alpha$ can be thought of as a finite collection, or 'chain,' of n-simplices in X with integer multiplicities, the coefficients n_α.

As one can see in the next figure, the boundary of the n-simplex $[v_0, \cdots, v_n]$ consists of the various $(n-1)$-dimensional simplices $[v_0, \cdots, \hat{v}_i, \cdots, v_n]$, where the 'hat' symbol $\hat{}$ over v_i indicates that this vertex is deleted from the sequence $v_0, \cdots, v_n$. In terms of chains, we might then wish to say that the boundary of $[v_0, \cdots, v_n]$ is the $(n-1)$-chain formed by the sum of the faces $[v_0, \cdots, \hat{v}_i, \cdots, v_n]$. However, it turns out to be better to insert certain signs and instead let the boundary of $[v_0, \cdots, v_n]$ be $\sum_i (-1)^i [v_0, \cdots, \hat{v}_i, \cdots, v_n]$. Heuristically, the signs are inserted to take orientations into account, so that all the faces of a simplex are coherently oriented, as indicated in the following figure:



$$\partial[v_0, v_1] = [v_1] - [v_0]$$

$$\partial[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]$$

$$\begin{aligned}\partial[v_0, v_1, v_2, v_3] = [v_1, v_2, v_3] &- [v_0, v_2, v_3] \\ &+ [v_0, v_1, v_3] - [v_0, v_1, v_2]\end{aligned}$$

In the last case, the orientations of the two hidden faces are also counterclockwise when viewed from outside the 3-simplex.

With this geometry in mind we define for a general Δ-complex X a **boundary homomorphism** $\partial_n : \Delta_n(X) \to \Delta_{n-1}(X)$ by specifying its values on basis elements:

$$\partial_n(\sigma_\alpha) = \sum_i (-1)^i \sigma_\alpha | [v_0, \cdots, \hat{v}_i, \cdots, v_n]$$

Note that the right side of this equation does indeed lie in $\Delta_{n-1}(X)$ since each restriction $\sigma_\alpha | [v_0, \cdots, \hat{v}_i, \cdots, v_n]$ is the characteristic map of an $(n-1)$-simplex of X.

Lemma 2.1. *The composition* $\Delta_n(X) \xrightarrow{\partial_n} \Delta_{n-1}(X) \xrightarrow{\partial_{n-1}} \Delta_{n-2}(X)$ *is zero.*

Proof: We have $\partial_n(\sigma) = \sum_i (-1)^i \sigma | [v_0, \cdots, \hat{v}_i, \cdots, v_n]$, and hence

$$\begin{aligned}\partial_{n-1}\partial_n(\sigma) &= \sum_{j<i} (-1)^i (-1)^j \sigma | [v_0, \cdots, \hat{v}_j, \cdots, \hat{v}_i, \cdots, v_n] \\ &\quad + \sum_{j>i} (-1)^i (-1)^{j-1} \sigma | [v_0, \cdots, \hat{v}_i, \cdots, \hat{v}_j, \cdots, v_n]\end{aligned}$$

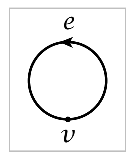
The latter two summations cancel since after switching i and j in the second sum, it becomes the negative of the first. $\square$

The algebraic situation we have now is a sequence of homomorphisms of abelian groups

$$\cdots \longrightarrow C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \longrightarrow \cdots \longrightarrow C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} 0$$

with $\partial_n\partial_{n+1} = 0$ for each n. Such a sequence is called a **chain complex**. Note that we have extended the sequence by a 0 at the right end, with $\partial_0 = 0$. From $\partial_n\partial_{n+1} = 0$ it follows that $\operatorname{Im}\partial_{n+1} \subset \operatorname{Ker}\partial_n$, where Im and Ker denote image and kernel. So we can define the n^{th} **homology group** of the chain complex to be the quotient group $H_n = \operatorname{Ker}\partial_n / \operatorname{Im}\partial_{n+1}$. Elements of $\operatorname{Ker}\partial_n$ are called **cycles** and elements of $\operatorname{Im}\partial_{n+1}$ are **boundaries**. Elements of H_n are cosets of $\operatorname{Im}\partial_{n+1}$, called **homology classes**. Two cycles representing the same homology class are said to be **homologous**. This means their difference is a boundary.

Returning to the case that $C_n = \Delta_n(X)$, the homology group $\operatorname{Ker}\partial_n / \operatorname{Im}\partial_{n+1}$ will be denoted $H_n^\Delta(X)$ and called the n^{th} **simplicial homology group** of X.



Example 2.2. $X = S^1$, with one vertex v and one edge e. Then $\Delta_0(S^1)$ and $\Delta_1(S^1)$ are both $\mathbb{Z}$ and the boundary map ∂_1 is zero since $\partial e = v - v$. The groups $\Delta_n(S^1)$ are 0 for $n \ge 2$ since there are no simplices in these dimensions. Hence

$$H_n^\Delta(S^1) \approx \begin{cases} \mathbb{Z} & \text{for } n = 0,1 \\ 0 & \text{for } n \ge 2 \end{cases}$$

This is an illustration of the general fact that if the boundary maps in a chain complex are all zero, then the homology groups of the complex are isomorphic to the chain groups themselves.

Example 2.3. $X = T$, the torus with the Δ-complex structure pictured earlier, having one vertex, three edges a, b, and c, and two 2-simplices U and L. As in the previous example, $\partial_1 = 0$ so $H_0^\Delta(T) \approx \mathbb{Z}$. Since $\partial_2 U = a + b - c = \partial_2 L$ and $\{a, b, a + b - c\}$ is a basis for $\Delta_1(T)$, it follows that $H_1^\Delta(T) \approx \mathbb{Z}\oplus\mathbb{Z}$ with basis the homology classes $[a]$ and $[b]$. Since there are no 3-simplices, $H_2^\Delta(T)$ is equal to $\operatorname{Ker}\partial_2$, which is infinite cyclic generated by $U - L$ since $\partial(pU + qL) = (p + q)(a + b - c) = 0$ only if $p = -q$. Thus

$$H_n^\Delta(T) \approx \begin{cases} \mathbb{Z}\oplus\mathbb{Z} & \text{for } n = 1 \\ \mathbb{Z} & \text{for } n = 0,2 \\ 0 & \text{for } n \ge 3 \end{cases}$$

Example 2.4. $X = \mathbb{R}P^2$, as pictured earlier, with two vertices v and w, three edges a, b, and c, and two 2-simplices U and L. Then $\operatorname{Im}\partial_1$ is generated by $w - v$, so $H_0^\Delta(X) \approx \mathbb{Z}$ with either vertex as a generator. Since $\partial_2 U = -a+b+c$ and $\partial_2 L = a-b+c$, we see that ∂_2 is injective, so $H_2^\Delta(X) = 0$. Further, $\operatorname{Ker}\partial_1 \approx \mathbb{Z}\oplus\mathbb{Z}$ with basis $a - b$ and c, and $\operatorname{Im}\partial_2$ is an index-two subgroup of $\operatorname{Ker}\partial_1$ since we can choose c and $a - b + c$

as a basis for $\operatorname{Ker}\partial_1$ and $a-b+c$ and $2c=(a-b+c)+(-a+b+c)$ as a basis for $\operatorname{Im}\partial_2$. Thus $H_1^\Delta(X)\approx\mathbb{Z}_2$.

Example 2.5. We can obtain a Δ-complex structure on S^n by taking two copies of Δ^n and identifying their boundaries via the identity map. Labeling these two n-simplices U and L, then it is obvious that $\operatorname{Ker}\partial_n$ is infinite cyclic generated by $U-L$. Thus $H_n^\Delta(S^n)\approx\mathbb{Z}$ for this Δ-complex structure on S^n. Computing the other homology groups would be more difficult.

Many similar examples could be worked out without much trouble, such as the other closed orientable and nonorientable surfaces. However, the calculations do tend to increase in complexity before long, particularly for higher-dimensional complexes.

Some obvious general questions arise: Are the groups $H_n^\Delta(X)$ independent of the choice of Δ-complex structure on X? In other words, if two Δ-complexes are homeomorphic, do they have isomorphic homology groups? More generally, do they have isomorphic homology groups if they are merely homotopy equivalent? To answer such questions and to develop a general theory it is best to leave the rather rigid simplicial realm and introduce the singular homology groups. These have the added advantage that they are defined for all spaces, not just Δ-complexes. At the end of this section, after some theory has been developed, we will show that simplicial and singular homology groups coincide for Δ-complexes.

Traditionally, simplicial homology is defined for **simplicial complexes**, which are the Δ-complexes whose simplices are uniquely determined by their vertices. This amounts to saying that each n-simplex has $n+1$ distinct vertices, and that no other n-simplex has this same set of vertices. Thus a simplicial complex can be described combinatorially as a set X_0 of vertices together with sets X_n of n-simplices, which are $(n+1)$-element subsets of X_0. The only requirement is that each $(k+1)$-element subset of the vertices of an n-simplex in X_n is a k-simplex, in X_k. From this combinatorial data a Δ-complex X can be constructed, once we choose a partial ordering of the vertices X_0 that restricts to a linear ordering on the vertices of each simplex in X_n. For example, we could just choose a linear ordering of all the vertices. This might perhaps involve invoking the Axiom of Choice for large vertex sets.

An exercise at the end of this section is to show that every Δ-complex can be subdivided to be a simplicial complex. In particular, every Δ-complex is then homeomorphic to a simplicial complex.

Compared with simplicial complexes, Δ-complexes have the advantage of simpler computations since fewer simplices are required. For example, to put a simplicial complex structure on the torus one needs at least 14 triangles, 21 edges, and 7 vertices, and for $\mathbb{R}\mathrm{P}^2$ one needs at least 10 triangles, 15 edges, and 6 vertices. This would slow down calculations considerably!

Singular Homology

A **singular n-simplex** in a space X is by definition just a map $\sigma:\Delta^n \to X$. The word 'singular' is used here to express the idea that σ need not be a nice embedding but can have 'singularities' where its image does not look at all like a simplex. All that is required is that σ be continuous. Let $C_n(X)$ be the free abelian group with basis the set of singular n-simplices in X. Elements of $C_n(X)$, called **n-chains**, or more precisely singular n-chains, are finite formal sums $\sum_i n_i\sigma_i$ for $n_i \in \mathbb{Z}$ and $\sigma_i:\Delta^n \to X$. A boundary map $\partial_n : C_n(X) \to C_{n-1}(X)$ is defined by the same formula as before:

$$\partial_n(\sigma) = \sum_i (-1)^i \sigma \,|\, [v_0, \cdots, \hat{v}_i, \cdots, v_n]$$

Implicit in this formula is the canonical identification of $[v_0, \cdots, \hat{v}_i, \cdots, v_n]$ with Δ^{n-1}, preserving the ordering of vertices, so that $\sigma \,|\, [v_0, \cdots, \hat{v}_i, \cdots, v_n]$ is regarded as a map $\Delta^{n-1} \to X$, that is, a singular $(n-1)$-simplex.

Often we write the boundary map ∂_n from $C_n(X)$ to $C_{n-1}(X)$ simply as ∂ when this does not lead to serious ambiguities. The proof of Lemma 2.1 applies equally well to singular simplices, showing that $\partial_n\partial_{n+1} = 0$ or more concisely $\partial^2 = 0$, so we can define the **singular homology group** $H_n(X) = \operatorname{Ker}\partial_n / \operatorname{Im}\partial_{n+1}$.

It is evident from the definition that homeomorphic spaces have isomorphic singular homology groups H_n, in contrast with the situation for H_n^Δ. On the other hand, since the groups $C_n(X)$ are so large, the number of singular n-simplices in X usually being uncountable, it is not at all clear that for a Δ-complex X with finitely many simplices, $H_n(X)$ should be finitely generated for all n, or that $H_n(X)$ should be zero for n larger than the dimension of X — two properties that are trivial for $H_n^\Delta(X)$.

Though singular homology looks so much more general than simplicial homology, it can actually be regarded as a special case of simplicial homology by means of the following construction. For an arbitrary space X, define the **singular complex** $S(X)$ to be the Δ-complex with one n-simplex Δ_σ^n for each singular n-simplex $\sigma:\Delta^n \to X$, with Δ_σ^n attached in the obvious way to the $(n-1)$-simplices of $S(X)$ that are the restrictions of σ to the various $(n-1)$-simplices in $\partial\Delta^n$. It is clear from the definitions that $H_n^\Delta(S(X))$ is identical with $H_n(X)$ for all n, and in this sense the singular homology group $H_n(X)$ is a special case of a simplicial homology group. One can regard $S(X)$ as a Δ-complex model for X, although it is usually an extremely large object compared to X.

Cycles in singular homology are defined algebraically, but they can be given a somewhat more geometric interpretation in terms of maps from finite Δ-complexes. To see this, note first that a singular n-chain ξ can always be written in the form $\sum_i \varepsilon_i\sigma_i$ with $\varepsilon_i = \pm 1$, allowing repetitions of the singular n-simplices σ_i. Given such an n-chain $\xi = \sum_i \varepsilon_i\sigma_i$, when we compute $\partial\xi$ as a sum of singular $(n-1)$-simplices with signs ± 1, there may be some *canceling pairs* consisting of two identical singular $(n-1)$-simplices with opposite signs. Choosing a maximal collection of such